Hengming Jing

Biomedical optics and computational imaging

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Education

Southern University of Science and Technology Sep 2024 – Jun 2027 (expected)
Master's degree in Electronic Science and Technology | GPA 3.44/4.00
Department of Biomedical Engineering | Advisor: Prof. Jianbo Tang

Hefei University of Technology Sep 2020 – Jun 2024
B.Eng. in Mechanical Engineering | GPA 3.62/4.30 | Top 10% (34/349, first six semesters)

Publications & Manuscripts

Journal articles

J. Tang, **H. Jing**, J. Yang, and W. Chen.
Dynamic optical coherence tomography: from qualitative to quantitative.
Chinese Journal of Lasers **53**(9), 0907106 (2026). Invited review. DOI

X. Zhai, **H. Jing**, H. Chen, S. Chang, and J. Tang.
3D reconstruction of murine hepatic vasculature via serial sectioning SS-OCT imaging.
Journal of Innovative Optical Health Sciences **19**(6), 2650006 (2026). DOI

F. Yang, C. Chen, L. Meng, **H. Jing**, X. Liu, W. Chen, S.-C. Chen, and J. Tang.
Learned feature-based clustering of brain tissue with multimodal optical coherence tomography and photoacoustic elastography.
Optics Letters **51**(3), 796–799 (2026). DOI

F. Yang, **H. Jing**, S. Chang, H. Sun, W. Chen, and J. Tang.
Multimodal label-free imaging of unprocessed biological tissues using ultraviolet photoacoustic elastography and optical coherence microscopy.
Optics Letters **50**(16), 5010–5013 (2025). DOI

Conference paper

J. Zhou, **H. Jing**, M. Li, and J. Tang.
OCT-based three-dimensional multi-parametric imaging of the cerebral microcirculation.
European Conferences on Biomedical Optics (ECBO), M1C.2 (2025). DOI

Preprint

T. Hussain, X. Zhai, **H. Jing**, M. A. Mohammed, and J. Tang.
Deep learning-based virtual staining in histopathology: from image similarity to clinical evidence.
SSRN, 7643703 (2026). DOI

Manuscripts in preparation

H. Jing, Z. Qin, and J. Tang.
Learning-based OCT angiography from M-mode scans. *In preparation.*

H. Jing, W. Li, J. Zhou, Z. Yang, and J. Tang.
Foundation-model adaptation and vascular graph analysis for 3D OCT angiography. *In preparation.*

H. Jing, H. Chen, S. Chang, X. Zhai, and J. Tang.
Automated deep-ultraviolet microscopy for label-free 3D tissue imaging. *In preparation.*

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Curriculum Vitae

Research Experience

Graduate research, Department of Biomedical Engineering, SUSTech

OCT Angiography from M-Mode OCT Mar 2026 – Present

- Formulated a learning-based approach to visualize microvasculature from M-mode OCT measurements.
- Evaluated small-vessel visibility and vascular continuity against the absolute complex difference baseline in 2D projections and 3D volumes.

3D OCTA Vessel Segmentation and Analysis Feb 2025 – Present

- Adapted the vesselFM foundation model to 3D OCT angiography for microvascular segmentation using manually refined pseudo-labels.
- Quantified vascular architecture in 13 OCTA volumes using centerline-based graphs to measure segment length, tortuosity, and apparent caliber. Code.

Automated Deep-UV Microscopy for 3D Tissue Imaging Apr 2025 – Mar 2026

- Established automated 255 nm deep-UV microscopy for label-free imaging of serial tissue sections.
- Combined extended-depth-of-field processing and illumination correction to preserve cellular and vascular detail in stitched tissue images.
- Reconstructed 3D mouse brain volumes and compared deep-UV tissue morphology with H&E histology.

Multimodal Label-Free Tissue Imaging Oct 2024 – Jan 2025

Second author; published in Optics Letters (2025)

- Built and aligned the photoacoustic subsystem and developed its acquisition controls.
- Conducted tissue imaging experiments to evaluate the complementary contrast of ultraviolet photoacoustic elastography and optical coherence microscopy.

Industry Research Experience

Software Development Intern, Raymind 2026

- Designed a cluster-based method to estimate leukocyte proportions from bright-field and fluorescence images.
- Reduced mean absolute error from 16.07 to 4.01 percentage points relative to the company baseline; evaluated 111 acquisition groups against same-day complete blood counts.

Presentations & Honors

Best Student Paper Award, PIBM 2026 Jul 2026

First-author oral presentation on 3D multiparametric imaging of the cerebral microvasculature at the International Conference on Photonics and Imaging in Biology and Medicine.

First-Class Graduate Scholarship, SUSTech 2024 – 2027

Undergraduate Scholarships, Hefei University of Technology 2020 – 2023

Second class (2020–21, 2022–23) and third class (2021–22).

Technical Skills

Computational imaging OCTA, image registration, 3D vessel segmentation, and quantitative vascular analysis.

Experimental imaging Optical system development and alignment, tissue microscopy, and in vivo mouse OCTA.

Software Python, PyTorch, MATLAB, PyQt, VTK, ParaView, SolidWorks, and Git.